Cell components

The nucleus

Sometimes called the heart (or nut) within the cell, the nucleus is a self-contained unit in which DNA, the organism's genetic material, is housed. DNA, deoxyribonucleic acid, is often called the blueprint material of the body because it contains all the coding information for making molecules required by the body. DNA is a nucleic acid and the most important part of chromosomes. It is the material from which genes are formed. The nucleolus (plural nucleoli) is a spherical body within the nucleus and manufactures RNA, ribonucleic acid. RNA is another nucleic acid and assists DNA in its roles. Nucleic acids perform two major functions. They pass on hereditary information and ensure the formation of key proteins.

Because the nucleus houses the information required for normal development, it 'oversees' and instructs how development will occur.

1 Match the words on the left with the correct meaning on the right by joining the corresponding arrows. Then write the correct information into a table.

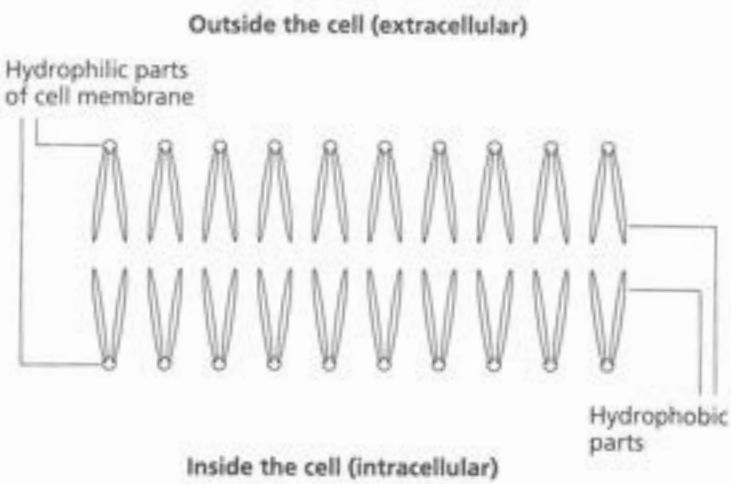
DNA	→	←	structures containing DNA
deoxyribose	→	←	unit containing a sugar, a base and a phosphate
gene	→	←	material which forms genes
chromosomes	→	←	sugar found in DNA
nucleotide	→	←	smallest unit of heredity
RNA	→	←	'little nucleus'
nucleolus	→	←	nucleic acid similar to DNA

Cyctoplasm

Cytoplasm consists of all of the cell substances outside the nucleus and inside the cell membrane. It is mainly a watery fluid (cytosol) with organelles suspended in it.

Cell membrane

Cell membrane acts as an envelope in which a cell's organelles, cytoplasm and nucleus are housed. It is also the gateway to transporting materials in and out of the cell. Substances inside the cell are intracellular and those outside the cell are extracellular. The membrane is made up of a double layer of phospholipids and cholesterol, positioned in a recognisable pattern. Phospholipids are lipids (fats/oils) having phosphate groups. Cholesterol is a complex alkanol (alcohol) that exists in all animal fats and oils. The head of each phospholipid molecule is hydrophilic, or 'water-loving', while the tails are hydrophobic, 'water-fearing'. In addition to phospholipids and cholesterol, protein molecules are embedded within the structure. The membrane also has many tiny openings (pores). The exact thickness of the membrane and type of protein embedded within it varies depending on the cell.



2 What are the major functions of the cell membrane?

3 What is meant by *hydrophilic*? **4** What is meant by *hydrophobic*? **5** What is a *lipid*?

6 What is a *pore*? **7** Describe the differences between a cell, an organelle and cytoplasm.

8 Describe the difference between the prefixes *intra* and *inter*.

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9 Discuss the importance of cytoplasm.

Vacuoles

Vacuoles are found within cells. The word vacuole means 'empty space' because that is how it appears under the microscope. Generally, plant cells have large vacuoles (sometimes taking up the majority of the cell) and animal cells have a greater number of much smaller vacuoles. Vacuoles contain water and dissolved substances.

10 Would a vacuole be considered an organelle? Explain your answer.

11 Suggest the outcome of one of a cell's organelles becoming damaged or dying.

Mitochondria

Mitochondria (singular – mitochondrion, from the Greek words meaning 'thread' and 'small piece of grain') can be round or long depending on the type of cell. They are very important in providing energy for the cell using cellular respiration. Mitochondria contain fats, proteins and enzymes.

12 Give a definition for *enzyme*.

One example of cellular respiration is the breakdown of glucose shown by the (much simplified) equation:

$$C_6H_{12}O_6 + 6\ O_2 + \text{enzymes} \rightarrow 6\ CO_2 + 6\ H_2O + \text{energy} + \text{enzymes}$$

Give the names of the following substances represented by chemical formulae:

13 $C_6H_{12}O_6$ _____

14 O_2 _____

15 CO_2 _____

16 H_2O _____

Endoplasmic reticulum

The endoplasm is the fluid part of the cytoplasm and a reticulum is a folded structure. The endoplasmic reticulum (ER) is a network of membranes that extends through the cytoplasm and is involved in the transport of materials within the cell. Many chemical reactions take place on the surface of the endoplasmic reticulum. Some parts are studded with ribosomes and are called the 'rough' ER. Areas not studded with ribosomes are called 'smooth' ER. *Ribosomes* are tiny granules that make proteins. Only rough ER is involved in making proteins. Smooth ER is involved in making transport vesicles.

Golgi apparatus (Golgi bodies)

This was discovered by Camillo Golgi in 1883. Golgi bodies are part of the endoplasmic reticulum and are flattened sacs. They are involved in the modification and transport of proteins and other substances.

Lysosomes

Lysosomes are small organelles containing various digestive enzymes. They break down worn-out cell parts and also any foreign materials that enter the cell.

Summarise the functions of the following organelles:

17 cell membrane

18 nucleus

19 mitochondrion

20 endoplasmic reticulum

21 Golgi apparatus

22 lysosomes

Research

23 Sometimes *lysosome* and *lysozyme* are confused. What is lysozyme?

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